# Commodity Derivatives Pricing Engine

Theoretical Foundations and Implementation

Schwartz-Smith Two-Factor Model · Kalman Filter Calibration
SABR Stochastic Volatility · Black-76 Pricing
Historical VaR/CVaR · Stress Testing


| | |
|---|---|
| **Author** | **Pablo M. Paniagua** |
| **Project** | QCEX — Quantum Commodity Exchange |
| **Commodities** | Cacao (ICE CC1) · Gas Natural TTF · Uranio U3O8 |
| **References** | Schwartz (1997) · Schwartz & Smith (2000)<br>Hagan et al. (2002) · Black (1976) |
| **Stack** | Python 3.11 · NumPy · SciPy · Streamlit · Plotly |

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## 1. Introduction and Motivation

This document provides the complete theoretical derivation underlying the **Commodity Derivatives Pricing Engine**, a production-grade quantitative system for pricing, calibrating, and risk-managing derivatives on physical commodities. The project is intentionally structured around a set of commodities that are analytically rich but underrepresented in standard quant portfolios.

### 1.1. Choice of Commodities

| Commodity | Exchange / Proxy | Unit | Key analytical feature |
|---|---|---|---|
| Cacao (Cocoa) | ICE Futures CC1 | USD/MT | Supply concentration, ENSO correlation, crisis |
| Gas Natural TTF | ICE TTF / Henry Hub NG | EUR/MWh | Stationarity seasonal, geopolitical basis risk, LN |
| Uranio U3O8 | OTC / Sprott URA ETF | USD/lb | OTC market, nuclear policy driven, thin smile |

Table 1: Commodities covered and their analytical characteristics

### 1.2. Architectural Overview

The system is structured in four modular layers:

1. **Data layer** (`data/fetcher.py`): yfinance as primary source with synthetic mean-reverting fallback, ensuring the app always runs.
2. **Models layer** (`models/`): Schwartz-Smith two-factor model with Kalman Filter; SABR stochastic vol with Hagan approximation; Black-76 option pricer with full Greeks.
3. **Analytics layer** (`analytics/risk.py`): Historical and parametric VaR/CVaR, vol cone, stress testing, drawdown analysis.
4. **Dashboard layer** (`app/`): Four-page Streamlit application with Plotly visualisations, live data feeds, and interactive calibration.

## 2. Theoretical Background: Commodity Price Dynamics

### 2.1. The Theory of Storage

The fundamental relationship between spot and futures prices in commodity markets was formalised by **Working (1949)** through the *theory of storage*. The cost-of-carry model states:

$$F(t,T) = S_t \cdot e^{(r+u-y)(T-t)} \tag{1}$$

where:

- $S_t$ is the spot price at time $t$
- $r$ is the risk-free interest rate
- $u$ is the cost of physical storage (annualised fraction)
- $y$ is the **convenience yield** — the implicit benefit of holding the physical commodity
- $T - t = \tau$ is the time to maturity

#### 2.1.1. Convenience Yield

The convenience yield $y$ is not directly observable. It is extracted by *bootstrapping* from the observed futures curve:

$$y(t,T) = r + u - \frac{1}{\tau}\ln\left(\frac{F(t,T)}{S_t}\right) \tag{2}$$

**Definition 2.1** (Market Structure). *When $y > r + u$, the futures curve is in **backwardation** ($F < S$), indicating tight physical supply: agents pay a premium for immediate delivery. When $y < r + u$, the market is in **contango** ($F > S$), implying storage costs dominate.*

**Remark 2.1.** *For commodities with strong seasonality (e.g. gas, cacao), the convenience yield $y(\tau)$ itself has term structure and is not constant. This motivates the stochastic models in Section 3.*

### 2.2. Stylised Facts of Commodity Prices

Commodity prices exhibit several empirical features that distinguish them from equity prices:

**F1.** **Mean reversion**: Prices tend toward a long-run equilibrium determined by production costs. Deviations induced by supply shocks are transient.

**F2.** **Seasonality**: Demand and supply are periodic (weather-driven for gas, harvest-driven for cacao).

**F3.** **Inventory effects**: Low inventories amplify price volatility and strengthen backwardation (Fama and French, 1987).

**F4.** **Negative skewness**: Supply disruptions cause sharp upward spikes. The return distribution has heavier left tails than normal when conditioned on negative outcomes for producers.

**F5.** **Stochastic volatility**: Implied vol smiles in commodity options are pronounced, requiring beyond-Black-Scholes models.

**F6.** **Imperfect futures correlation**: Near-term and long-term futures prices are not perfectly correlated, invalidating single-factor models.

## 3. The Schwartz-Smith Two-Factor Model

### 3.1. Motivation for Two Factors

**Schwartz (1997)** proposed a one-factor mean-reverting model for commodity prices. However, Schwartz and Smith (2000) demonstrated empirically that a single latent factor implies perfect correlation across the entire futures term structure, a hypothesis strongly rejected by the data. The two-factor model resolves this by separating the price process into:

- A **short-term factor** $\chi_t$: transient deviations from equilibrium (weather shocks, inventory surprises, geopolitical events)
- A **long-term factor** $\xi_t$: the equilibrium price level, evolving slowly with production costs, energy transition, and long-run supply/demand

$$\ln S_t = \chi_t + \xi_t \tag{3}$$

### 3.2. Physical Measure Dynamics

Under the real-world probability measure $\mathbb{P}$:

$$\mathrm{d}\chi_t = -\kappa\,\chi_t\,\mathrm{d}t + \sigma_\chi\,\mathrm{d}W_\chi^{\mathbb{P}} \tag{4}$$

$$\mathrm{d}\xi_t = \mu_\xi\,\mathrm{d}t + \sigma_\xi\,\mathrm{d}W_\xi^{\mathbb{P}} \tag{5}$$

$$\mathrm{d}W_\chi^{\mathbb{P}}\,\mathrm{d}W_\xi^{\mathbb{P}} = \rho\,\mathrm{d}t \tag{6}$$

| Parameter | Description | Interpretation |
|---|---|---|
| $\kappa > 0$ | Mean-reversion speed | Rate at which $\chi_t$ returns to zero. Half-life $= \ln 2/\kappa$ |
| $\mu_\xi$ | Long-term drift | Expected appreciation of equilibrium price |
| $\sigma_\chi$ | Short-term vol | Volatility of transient shocks |
| $\sigma_\xi$ | Long-term vol | Volatility of equilibrium level |
| $\rho \in (-1, 1)$ | Correlation | Between short-term and long-term Brownian motions |

Table 2: Schwartz-Smith model parameters under $\mathbb{P}$

#### 3.2.1. Factor Properties

Equation (4) is an **Ornstein-Uhlenbeck** (OU) process. By Itô's formula:

$$\chi_t = \chi_0 e^{-\kappa t} + \sigma_\chi \int_0^t e^{-\kappa(t-s)}\,\mathrm{d}W_\chi^{\mathbb{P}}(s) \tag{7}$$

which implies:

$$\mathbb{E}[\chi_t] = \chi_0\,e^{-\kappa t} \xrightarrow{t\to\infty} 0 \tag{8}$$

$$\mathrm{Var}[\chi_t] = \frac{\sigma_\chi^2}{2\kappa}\left(1 - e^{-2\kappa t}\right) \xrightarrow{t\to\infty} \frac{\sigma_\chi^2}{2\kappa} \tag{9}$$

The **half-life** of the short-term factor is $h = \ln 2/\kappa$. For our calibrated commodities:

- **Cacao** ($\kappa = 0.80$): half-life $\approx$ 10.4 months
- **Gas TTF** ($\kappa = 2.10$): half-life $\approx$ 4.0 months
- **Uranium** ($\kappa = 0.30$): half-life $\approx$ 27.6 months

Equation (5) is a **Brownian motion with drift**, the long-term permanent component. Unlike $\chi_t$, it does not mean-revert.

### 3.3. Risk-Neutral Measure

Introducing constant market prices of risk $\lambda_\chi$ (short-term) and $\lambda_\xi$ (long-term), by Girsanov's theorem:

$$\mathrm{d}W_\chi^{\mathbb{Q}} = \mathrm{d}W_\chi^{\mathbb{P}} + \frac{\lambda_\chi}{\sigma_\chi}\,\mathrm{d}t \tag{10}$$

$$\mathrm{d}W_\xi^{\mathbb{Q}} = \mathrm{d}W_\xi^{\mathbb{P}} + \frac{\lambda_\xi}{\sigma_\xi}\,\mathrm{d}t \tag{11}$$

Under $\mathbb{Q}$ the dynamics become:

$$\mathrm{d}\chi_t = \left(-\kappa\chi_t - \lambda_\chi\right)\,\mathrm{d}t + \sigma_\chi\,\mathrm{d}W_\chi^{\mathbb{Q}} \tag{12}$$

$$\mathrm{d}\xi_t = \left(\mu_\xi - \lambda_\xi\right)\,\mathrm{d}t + \sigma_\xi\,\mathrm{d}W_\xi^{\mathbb{Q}} \tag{13}$$

The short-term factor now mean-reverts to $-\lambda_\chi/\kappa$ under $\mathbb{Q}$, encoding the market price of short-term supply risk.

### 3.4. Futures Pricing Formula

**Theorem 3.1** (Schwartz-Smith Futures Price)**.** *Under the two-factor model, the time-t futures price for delivery at T is:*

$$\ln F(t,T) = e^{-\kappa\tau}\chi_t + \xi_t + A(\tau) \tag{14}$$

*where $\tau = T - t$ and the deterministic correction is:*

$$A(\tau) = \underbrace{(\mu_\xi - \lambda_\xi)\tau}_{\textit{risk-adj. drift}} - \underbrace{\frac{1-e^{-\kappa\tau}}{\kappa}\lambda_\chi}_{\textit{S-T risk prem.}} + \underbrace{\frac{\sigma_\xi^2\tau}{2}}_{\textit{Jensen (L-T)}} + \underbrace{\frac{\sigma_\chi^2(1-e^{-2\kappa\tau})}{4\kappa}}_{\textit{Jensen (S-T)}} + \underbrace{\frac{\rho\sigma_\chi\sigma_\xi(1-e^{-\kappa\tau})}{\kappa}}_{\textit{cross-factor}} \tag{15}$$

*Sketch.* By the no-arbitrage condition, $F(t,T) = \mathbb{E}^{\mathbb{Q}}[S_T \mid \mathcal{F}_t]$. Since $\ln S_T = \chi_T + \xi_T$ is Gaussian under $\mathbb{Q}$ (both factors follow linear SDEs), we apply the lognormal moment generating function:

$$\mathbb{E}^{\mathbb{Q}}[e^X] = e^{\mathbb{E}^{\mathbb{Q}}[X] + \frac{1}{2}\operatorname{Var}^{\mathbb{Q}}[X]}$$

Computing $\mathbb{E}^{\mathbb{Q}}[\chi_T \mid \mathcal{F}_t]$, $\mathbb{E}^{\mathbb{Q}}[\xi_T \mid \mathcal{F}_t]$, and the conditional variances and covariance yields (14)–(15). □

#### 3.4.1. Key Structural Properties of $A(\tau)$

As $\tau \to \infty$: $e^{-\kappa\tau} \to 0$, so $\ln F(t,T) \to \xi_t + A(\tau)$ — long-dated futures depend *only* on the permanent factor $\xi_t$.

As $\tau \to 0$: $F(t,T) \to S_t$ — no-arbitrage boundary condition.

The slope of the log-futures curve at short maturities is:

$$\left.\frac{\partial \ln F}{\partial \tau}\right|_{\tau=0} = -\kappa\chi_t + (\mu_\xi - \lambda_\xi) - \lambda_\chi + \sigma_\xi^2/2 + \rho\sigma_\chi\sigma_\xi$$

A large negative $\chi_t$ (supply shock pushing prices above equilibrium) induces steep backwardation in the short end of the curve.

### 3.5. Convenience Yield Extraction

From the model, the implied convenience yield consistent with (14) is:

$$y(\tau) = r + u - \frac{A(\tau)}{\tau} - \frac{e^{-\kappa\tau}-1}{\tau}\chi_t + \frac{\xi_t - \ln S_t}{\tau} \tag{16}$$

In the implementation, we use the simpler observable formula (2) bootstrapped from market futures prices.

### 3.6. Exact Discretisation and State-Space Form

Let $\Delta t$ be the observation interval. Define the state vector $\mathbf{x}_t = (\chi_t, \xi_t)^\top$. The exact discretisation of the joint SDE system is:

$$\mathbf{x}_t = \mathbf{F}\,\mathbf{x}_{t-1} + \boldsymbol{\eta}_t, \qquad \boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) \tag{17}$$

$$\mathbf{F} = \begin{pmatrix} e^{-\kappa\Delta t} & 0 \\ 0 & 1 \end{pmatrix}, \qquad \mathbf{Q} = \begin{pmatrix} Q_{11} & Q_{12} \\ Q_{12} & Q_{22} \end{pmatrix} \tag{18}$$

$$Q_{11} = \frac{\sigma_\chi^2}{2\kappa}\left(1 - e^{-2\kappa\Delta t}\right) \tag{19}$$

$$Q_{22} = \sigma_\xi^2\,\Delta t \tag{20}$$

$$Q_{12} = \frac{\rho\sigma_\chi\sigma_\xi}{\kappa}\left(1 - e^{-\kappa\Delta t}\right) \tag{21}$$

The observation equation for $n$ futures contracts with maturities $\tau_1, \ldots, \tau_n$ is:

$$\mathbf{y}_t = \mathbf{H}\,\mathbf{x}_t + \mathbf{d} + \boldsymbol{\varepsilon}_t, \qquad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}) \tag{22}$$

where $\mathbf{y}_t = (\ln F(t, T_1), \ldots, \ln F(t, T_n))^\top$ and:

$$\mathbf{H} = \begin{pmatrix} e^{-\kappa\tau_1} & 1 \\ \vdots & \vdots \\ e^{-\kappa\tau_n} & 1 \end{pmatrix}, \qquad \mathbf{d} = \begin{pmatrix} A(\tau_1) \\ \vdots \\ A(\tau_n) \end{pmatrix}, \qquad \mathbf{R} = s^2\mathbf{I}_n \tag{23}$$

with $s^2$ a homoscedastic measurement noise variance.

## 4. Kalman Filter and Maximum Likelihood Calibration

### 4.1. Kalman Filter Recursion

The linear Gaussian state-space system (17)–(22) admits the exact **Kalman Filter** for optimal (minimum mean-squared error) estimation of latent states $\{\chi_t, \xi_t\}$ from observable log-futures prices.

#### 4.1.1. Initialisation

The short-term factor $\chi_t$ has a known stationary distribution:

$$\chi_0 \sim \mathcal{N}\left(0, \frac{\sigma_\chi^2}{2\kappa}\right)$$

The long-term factor $\xi_t$ has a diffuse (uninformative) prior with large variance.

#### 4.1.2. Prediction Step

$$\hat{\mathbf{x}}_{t|t-1} = \mathbf{F}\,\hat{\mathbf{x}}_{t-1|t-1} \tag{24}$$

$$\mathbf{P}_{t|t-1} = \mathbf{F}\,\mathbf{P}_{t-1|t-1}\,\mathbf{F}^\top + \mathbf{Q} \tag{25}$$

#### 4.1.3. Innovation

$$\mathbf{v}_t = \mathbf{y}_t - \mathbf{H}\,\hat{\mathbf{x}}_{t|t-1} - \mathbf{d} \tag{26}$$

$$\mathbf{S}_t = \mathbf{H}\,\mathbf{P}_{t|t-1}\,\mathbf{H}^\top + \mathbf{R} \tag{27}$$

#### 4.1.4. Update Step (Kalman Gain)

$$\mathbf{K}_t = \mathbf{P}_{t|t-1}\,\mathbf{H}^\top\,\mathbf{S}_t^{-1} \tag{28}$$

$$\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + \mathbf{K}_t\,\mathbf{v}_t \tag{29}$$

$$\mathbf{P}_{t|t} = (\mathbf{I} - \mathbf{K}_t\,\mathbf{H})\,\mathbf{P}_{t|t-1} \tag{30}$$

### 4.2. Log-Likelihood and MLE Calibration

The innovations $\mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{S}_t)$ (under a well-specified model). The **prediction error decomposition** of the log-likelihood is:

$$\ell(\boldsymbol{\theta}) = -\frac{1}{2}\sum_{t=1}^{T}\Big[\ln|\mathbf{S}_t| + \mathbf{v}_t^\top\,\mathbf{S}_t^{-1}\,\mathbf{v}_t + n\ln(2\pi)\Big] \tag{31}$$

where $\boldsymbol{\theta} = (\kappa, \mu_\xi, \sigma_\chi, \sigma_\xi, \rho, \lambda_\chi, \lambda_\xi, s)$ is the parameter vector.
Maximum likelihood estimates are obtained by solving:

$$\hat{\boldsymbol{\theta}}_{\text{MLE}} = \arg\max_{\boldsymbol{\theta}}\ \ell(\boldsymbol{\theta})$$

via **L-BFGS-B** (gradient-based, bounded) with multiple random restarts to mitigate local optima.

#### 4.2.1. Identification

Parameters $\mu_\xi$ and $\lambda_\xi$ are not separately identified from cross-sectional futures data alone (only their difference $\mu_\xi - \lambda_\xi$ enters $A(\tau)$). Identification requires either time-series variation in the level of the forward curve, or imposing restrictions on the market price of risk.

### 4.3. Model Diagnostics

A correctly specified model produces *whitened innovations*:

- $\mathbb{E}[\mathbf{v}_t] \approx \mathbf{0}$: no systematic pricing bias
- $\text{AC}(\mathbf{v}_t, \mathbf{v}_{t-k}) \approx 0$ for $k \geq 1$: no serial correlation (Ljung-Box test)
- Approximate normality (Jarque-Bera test)

## 5. Monte Carlo Simulation

### 5.1. Path Simulation Algorithm

Given calibrated parameters $\hat{\boldsymbol{\theta}}$ and filtered states $(\hat{\chi}_T, \hat{\xi}_T)$ at the last observation date, future spot paths are simulated under $\mathbb{P}$:

**Step 1.** Compute Cholesky decomposition of the $2 \times 2$ correlation matrix:

$$\mathbf{\Sigma} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} = \mathbf{L}\mathbf{L}^\top$$

**Step 2.** At each step $t = 1, \ldots, N$, draw correlated increments:

$$(Z_\chi, Z_\xi)^\top = \mathbf{L}\,\mathbf{Z}, \quad \mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_2)$$

**Step 3.** Update factors using exact discretisation for $\chi$ and Euler-Maruyama for $\xi$:

$$\chi_t = \chi_{t-1}\, e^{-\kappa \Delta t} + \sigma_\chi \sqrt{\frac{1 - e^{-2\kappa\Delta t}}{2\kappa}}\, Z_\chi \tag{32}$$

$$\xi_t = \xi_{t-1} + \mu_\xi \Delta t + \sigma_\xi \sqrt{\Delta t}\, Z_\xi \tag{33}$$

**Step 4.** Reconstruct spot: $S_t = \exp(\chi_t + \xi_t)$

The exact discretisation for $\chi_t$ (step 3) avoids discretisation bias that would occur with Euler-Maruyama on the OU process.

### 5.2. Monte Carlo Option Pricing

For a European call on spot with strike $K$, maturity $T$, and risk-free rate $r$:

$$C_{\text{MC}} = e^{-rT} \frac{1}{N_{\text{paths}}} \sum_{i=1}^{N_{\text{paths}}} \max(S_T^{(i)} - K,\ 0) \tag{34}$$

The standard error is:

$$\widehat{\text{SE}} = e^{-rT} \frac{\hat{\sigma}_{\text{payoff}}}{\sqrt{N_{\text{paths}}}}$$

yielding a 95% confidence interval of width $2 \times 1.96\,\widehat{\text{SE}}$.

## 6. Black-76 Model for Options on Futures

### 6.1. Assumptions

Black (1976) extends the Black-Scholes framework to options whose underlying is a futures contract $F(t, T_F)$ rather than a spot asset. The key simplification: futures require no initial investment, so no cost-of-carry term appears.

Under $\mathbb{Q}$, the futures price follows:

$$\mathrm{d}F = \sigma F\ \mathrm{d}W^{\mathbb{Q}}$$

### 6.2. Pricing Formulas

$$C = e^{-r\tau}\left[F\,N(d_1) - K\,N(d_2)\right] \tag{35}$$

$$P = e^{-r\tau}\left[K\,N(-d_2) - F\,N(-d_1)\right] \tag{36}$$

$$d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau} \tag{37}$$

where $N(\cdot)$ is the standard normal CDF.

### 6.3. The Greeks

| Greek | Formula (call) | Financial meaning | Units |
|---|---|---|---|
| $\Delta$ | $e^{-r\tau}N(d_1)$ | Sensitivity to $F$ | Dimensionless |
| $\Gamma$ | $\dfrac{e^{-r\tau}\phi(d_1)}{F\sigma\sqrt{\tau}}$ | Convexity w.r.t. $F$ | per unit of $F$ |
| $\nu$ (Vega) | $F\,e^{-r\tau}\phi(d_1)\sqrt{\tau}$ | Sensitivity to $\sigma$ | per unit of vol |
| $\Theta$ | $-\dfrac{F\,e^{-r\tau}\phi(d_1)\sigma}{2\sqrt{\tau}} - rKe^{-r\tau}N(d_2)$ | Time decay (annual) | per year |
| *Second-order Greeks (computed via finite differences in the implementation)* | | | |
| Vanna | $\partial^2 V/\partial F\,\partial\sigma$ | Mixed $F$-vol sensitivity | Crucial for barrier/digital opt |
| Volga | $\partial^2 V/\partial\sigma^2$ | Vol convexity; captures smile risk | Vol smile curvature premium |

Table 3: Black-76 Greeks for call options. $\phi(\cdot)$ denotes the standard normal PDF.

#### 6.3.1. Implied Volatility Inversion

Given a market option price $C_{\text{mkt}}$, the implied vol $\hat{\sigma}$ satisfies:

$$C_{\text{B76}}(F, K, r, \hat{\sigma}, \tau) = C_{\text{mkt}}$$

Inverted via **Newton-Raphson**: starting from $\sigma_0 = 0.30$, iterate:

$$\sigma_{n+1} = \sigma_n - \frac{C_{\text{B76}}(\sigma_n) - C_{\text{mkt}}}{\nu(\sigma_n)}$$

convergence guaranteed in $O(10)$ iterations due to the monotonicity of $C$ in $\sigma$ and the convexity of the vega function.

## 7. SABR Stochastic Volatility Model

### 7.1. Motivation

Black-76 assumes constant volatility, producing a *flat vol surface*: all strikes and maturities are priced with the same $\sigma$. In reality, commodity options exhibit:

- **Negative skew**: OTM puts are more expensive than OTM calls (supply risk)
- **Smile curvature**: Both wings trade above ATM vol (high vol-of-vol)
- **Term structure of smile**: The shape evolves with maturity

The **SABR model** (Hagan, Kumar, Lesniewski and Woodward, 2002) addresses this with a two-dimensional stochastic process for the forward $F$ and its volatility $\sigma$.

### 7.2. Dynamics

$$\mathrm{d}F = \sigma F^{\beta}\ \mathrm{d}W_1 \tag{38}$$
$$\mathrm{d}\sigma = \alpha\,\sigma\ \mathrm{d}W_2 \tag{39}$$
$$\mathrm{d}W_1\ \mathrm{d}W_2 = \rho\ \mathrm{d}t \tag{40}$$

| Param. | Role | Commodity intuition |
|---|---|---|
| $\alpha > 0$ | Initial vol level | Scales the smile vertically; calibrates to ATM vol |
| $\beta \in [0,1]$ | CEV elasticity | $\beta = 0$: normal; $\beta = 1$: lognormal; $\beta = 0.5$: standard for commodities |
| $\rho \in (-1,1)$ | Skew driver | $\rho < 0$: negative skew (puts expensive). Supply risk in cacao, gas |
| $\nu > 0$ | Vol of vol | Smile curvature. High $\nu$ in energy (spikes); lower in OTC uranium |

Table 4: SABR model parameters and their economic interpretation

**Remark 7.1.** *The SABR model has no mean reversion in $\sigma$, unlike the Heston model. This is intentional: it prioritises tractability of the smile approximation over long-term vol dynamics.*

### 7.3. Hagan (2002) Implied Volatility Approximation

The exact SABR model has no closed-form option price. Hagan et al. derived a **singular perturbation expansion** in the vol-of-vol parameter $\alpha$, yielding:

$$\sigma_B(K,T) \approx \frac{\alpha}{D(K)} \cdot \frac{z}{\chi(z)} \cdot C(T) \tag{41}$$

where:

$$D(K) = (FK)^{(1-\beta)/2}\left[1 + \frac{(1-\beta)^2}{24}\ln^2\frac{F}{K} + \frac{(1-\beta)^4}{1920}\ln^4\frac{F}{K}\right] \tag{42}$$
$$z = \frac{\nu}{\alpha}(FK)^{(1-\beta)/2}\ln\frac{F}{K} \tag{43}$$
$$\chi(z) = \ln\left[\frac{\sqrt{1-2\rho z+z^2}+z-\rho}{1-\rho}\right] \tag{44}$$
$$C(T) = 1 + T\left[\frac{(1-\beta)^2\alpha^2}{24(FK)^{1-\beta}} + \frac{\rho\beta\nu\alpha}{4(FK)^{(1-\beta)/2}} + \frac{\nu^2(2-3\rho^2)}{24}\right] \tag{45}$$

#### 7.3.1. ATM Limit ($F = K$)

Taking the limit $K \to F$ (L'Hôpital's rule gives $z/\chi(z) \to 1$):

$$\sigma_{\mathrm{ATM}}(T) \approx \frac{\alpha}{F^{1-\beta}}\left[1 + T\left(\frac{(1-\beta)^2\alpha^2}{24F^{2(1-\beta)}} + \frac{\rho\beta\nu\alpha}{4F^{1-\beta}} + \frac{\nu^2(2-3\rho^2)}{24}\right)\right] \tag{46}$$

At $T = 0$: $\sigma_{\mathrm{ATM}} \approx \alpha/F^{1-\beta}$, so $\alpha$ is directly the ATM vol rescaled by the forward.

## 7.4. Smile Interpretation

### 7.4.1. Skew

The first-order slope of the smile around ATM is controlled by $\rho$:

$$\left.\frac{\partial \sigma_B}{\partial K}\right|_{K=F} \approx -\frac{\rho\nu}{2F^{(1-\beta)/2}}$$

Negative $\rho$ induces negative skew (downward-sloping smile from left to right): the market assigns higher implied vol to downside strikes (OTM puts). This is the dominant effect for commodities with supply concentration risk (cacao, gas in winter).

### 7.4.2. Smile Curvature

The second-order curvature ("smile" or "butterfly") is driven by $\nu$:

$$\left.\frac{\partial^2 \sigma_B}{\partial K^2}\right|_{K=F} \propto \nu^2$$

High $\nu$ (as in gas TTF: $\nu = 0.85$) produces pronounced wings. Low $\nu$ (uranium: $\nu = 0.42$) produces a flatter, more symmetric profile.

## 7.5. SABR Calibration

With $\beta$ fixed (typically $\beta = 0.5$ for commodities), the remaining three parameters $(\alpha, \rho, \nu)$ are calibrated per maturity slice by **weighted least squares** on implied vol:

$$(\hat{\alpha}, \hat{\rho}, \hat{\nu}) = \arg\min_{\alpha,\rho,\nu} \sum_{i=1}^{n} w_i \left[\sigma_B(K_i, T; \alpha, \rho, \nu) - \sigma_i^{\text{mkt}}\right]^2 \tag{47}$$

Solved via multi-start **L-BFGS-B**. Calibration is performed independently for each maturity slice; the resulting parameter surface $\{(\hat{\alpha}(T), \hat{\rho}(T), \hat{\nu}(T))\}$ describes the full vol surface.

### 7.5.1. Calibration Results (Cacao, $T = 6M$)

| | $\alpha$ | $\rho$ | $\nu$ | **RMSE (bp)** |
|---|---|---|---|---|
| **Prior (calibrated)** | 0.52 | −0.45 | 0.68 | — |
| **Fitted (synthetic data)** | 0.80 | −0.50 | 0.56 | 33.6 |

Table 5: SABR calibration to synthetic market smile (11 strikes, T=6M, $\beta = 0.5$)

## 7.6. Vol Surface and Term Structure

The full **implied vol surface** is the map $(K, T) \mapsto \sigma_B(K, T)$ computed from (41) for a grid of strikes $K \in [0.70F, 1.30F]$ and maturities $T \in \{1M, 2M, \ldots, 2Y\}$.

Key cross-sectional features:

- **Term structure of ATM vol**: $\sigma_{\text{ATM}}(T)$ typically increases with $T$ ("normal" vol term structure) but can invert during supply crises (backwardation of vol)
- **Skew term structure**: The skew $|\rho|$ tends to decrease with maturity as long-term supply becomes more elastic
- **Smile decays**: At very long maturities, $C(T)$ corrections become large and the Hagan approximation may fail (arbitrage in wings for $T > 3Y$)

#### 7.6.1. SABR Limitations

**L1. Absorbed probability mass**: The SABR process can hit zero in finite time (when $\beta < 1$). For long maturities, the approximation (41) can violate calendar spread arbitrage.

**L2. Static per-slice**: Calibrating independently per slice does not ensure a consistent dynamic model for volatility. Extensions include SABR-LSV (Local Stochastic Volatility).

**L3. Thin markets**: For uranium (OTC), the smile data is sparse and noisy. The calibration is more uncertain; $\nu$ estimation requires wide bid-ask spreads.

## 8. Risk Analytics

### 8.1. Value at Risk

**Definition 8.1** (Historical VaR). *The $(1-p)$-quantile of the empirical distribution of scaled log-returns:*

$$VaR_{p,h} = -Q_p\left(\{r_i\sqrt{h}\}_{i=1}^{T}\right) \tag{48}$$

*where $r_i = \ln(S_i/S_{i-1})$ are daily log-returns and $h$ is the holding period in days.*

**Definition 8.2** (Parametric (Gaussian) VaR).

$$VaR_{p,h}^{Gauss} = \hat{\sigma}\,\Phi^{-1}(1-p)\,\sqrt{h} \tag{49}$$

*where $\hat{\sigma}$ is the sample standard deviation of daily log-returns.*

**Definition 8.3** (Expected Shortfall (CVaR)).

$$CVaR_{p,h} = -\mathbb{E}\left[r\sqrt{h} \;\middle|\; r\sqrt{h} < -VaR_{p,h}\right] \tag{50}$$

*CVaR is a coherent risk measure (Artzner et al., 1999) and the regulatory standard under FRTB (Basel III/IV).*

**Remark 8.1.** *Commodity returns exhibit excess kurtosis (fat tails), so Gaussian VaR typically* underestimates *true risk. Historical VaR is preferred for commodities.*

### 8.2. Stress Testing

The stress scenarios implemented are commodity-specific, calibrated to historical crisis events:

| Commodity | Scenario | Shock (%) | Mechanism |
|---|---|---|---|
| Cacao | El Niño severo | −45 | Drought in West Africa; crop failure |
| Cacao | Fungal disease | −35 | Swollen shoot destroys 30% harvest |
| Gas TTF | Invierno frío extremo | +80 | HDD +40%; heating demand surge |
| Gas TTF | Resolución conflicto | −50 | Russian pipeline reopening; supply glut |
| Uranio | Kazatomprom recorte | +50 | Kazakhstan cuts output 20% (43% global share) |
| Uranio | Accidente nuclear | −40 | Level 4–5 event; reactor shutdowns |

Table 6: Selected stress scenarios (full list in `analytics/risk.py`)

### 8.3. Volatility Cone

The **volatility cone** provides a distributional view of realised volatility across measurement horizons:

$$\hat{\sigma}_w(t) = \sqrt{252}\,\hat{\sigma}\big(\{r_i\}_{i=t-w+1}^{t}\big) \tag{51}$$

For each window $w \in \{5, 10, 21, 63, 126, 252\}$, we compute the empirical percentiles $\{P_{10}, P_{25}, P_{50}, P_{75}, P_{90}\}$ and the current value. This identifies whether current vol is historically cheap or expensive — a key input to options strategies.

## 9. System Architecture and Implementation

### 9.1. Module Structure

| Module | Key functions | Dependencies |
|---|---|---|
| `models/schwartz_smith.py` | `kalman_filter`, `calibrate`, `simulate_paths` | NumPy, SciPy |
| `models/sabr.py` | `SABRModel`, `sabr_implied_vol`, `sabr_greeks` | NumPy, SciPy |
| `analytics/risk.py` | `historical_var`, `stress_table`, `vol_cone` | NumPy, Pandas |
| `data/fetcher.py` | `fetch_prices`, `build_term_structure` | yfinance, Pandas |
| `app/main.py` | Streamlit entry point | Streamlit, Plotly |

Table 7: Module structure and key components

### 9.2. Data Architecture

The data layer uses a **two-tier fallback** strategy:

**T1. Primary**: `yfinance` REST API (CC=F cacao, NG=F gas, URA uranium ETF)

**T2. Fallback**: Synthetic data generated via the calibrated Schwartz 1-factor OU process, ensuring the app always runs in offline or demo environments

All data fetches are cached at the Streamlit layer with a 1-hour TTL.

### 9.3. Calibration Performance

### 9.4. Key Design Decisions

#### 9.4.1. Why Schwartz-Smith vs Single-Factor?

Single-factor models produce perfect cross-sectional correlation in the futures curve: a property strongly rejected empirically. Schwartz-Smith (2000) tested this on crude oil, copper, and gold, finding significant two-factor structure. For cacao (where weather-driven short-term spikes are largely uncorrelated with the long-run equilibrium set by production costs), the two-factor decomposition is particularly valuable.

| **Model** | **Time (s)** | **RMSE** | **Method** |
|---|---|---|---|
| Kalman Filter (single pass, $T = 260$, $n = 6$) | $< 0.01$ | — | Exact recursion |
| Schwartz-Smith MLE (5 restarts) | $\approx 48$ | — | L-BFGS-B |
| SABR calibration (11 strikes) | 0.20 | 33.6 bp | Multi-start L-BFGS-B |
| Black-76 implied vol (Newton-Raphson) | $< 0.001$ | $< 10^{-8}$ | 10 iterations |
| Monte Carlo option price (50,000 paths) | $\approx 2$ | $\pm 0.0004$ | Cholesky corr. |

Table 8: Computational performance benchmarks

#### 9.4.2. Why $\beta = 0.5$ in SABR?

The CEV exponent $\beta$ controls the relationship between the forward level and its instantaneous vol. Pure log-normal ($\beta = 1$) is standard for equity and FX, but it implies volatility grows linearly with price — problematic for commodities that can spike to extreme levels. Normal ($\beta = 0$) implies negative prices are possible. $\beta = 0.5$ (geometric Brownian motion with stochastic vol) is the industry standard for commodity options desks: it keeps prices positive while allowing moderate vol-level dependence.

#### 9.4.3. Why Historical VaR over Parametric?

Commodity returns have excess kurtosis and negative skewness, violating the normality assumption of parametric VaR. For cacao, the 2023–24 crisis produced daily returns in the $\pm 8\%$ range, far outside a Gaussian $\pm 3\sigma$ band. Historical VaR captures this empirically without distributional assumptions.

## 10. Commodity-Specific Analysis

### 10.1. Cacao (Cocoa)

#### 10.1.1. Market Structure

The global cocoa market is characterised by extreme geographic concentration: Ivory Coast and Ghana together account for approximately 65% of world production, creating a structurally supply-inelastic commodity vulnerable to weather shocks.

#### 10.1.2. ENSO and Seasonal Dynamics

El Niño Southern Oscillation (ENSO) episodes correlate strongly with reduced rainfall in West Africa. The 2023–24 El Niño triggered a crop failure that drove spot prices from $\approx \$3,000/\text{MT}$ to over $\$12,000/\text{MT}$ — a historical anomaly that produced the high calibrated volatility $\sigma_\chi = 0.38$ and the strong negative SABR skew $\rho = -0.45$.

#### 10.1.3. Calibrated Parameters

### 10.2. Gas Natural TTF

#### 10.2.1. Post-2022 Regime

Following the energy crisis triggered by the reduction of Russian pipeline supplies to Europe in 2022, TTF gas volatility shifted to a structurally higher regime. Historical vol moved from

| Parameter | Value (calibrated) |
|---|---|
| $\kappa$ (mean-reversion) | 0.80 $\text{yr}^{-1} \Rightarrow$ half-life $\approx$ 10.4 months |
| $\sigma_\chi$ (S-T vol) | 38% p.a. (crisis-calibrated) |
| $\sigma_\xi$ (L-T vol) | 18% p.a. |
| $\rho$ (correlation) | $-0.25$ |
| SABR $\rho$ (skew) | $-0.45$ (strong supply risk skew) |
| SABR $\nu$ (smile) | 0.68 |

$\approx$ 30% to peaks above 200% annualised, with the SABR vol-of-vol parameter $\nu = 0.85$ reflecting persistent spike risk.

#### 10.2.2. Seasonality

Gas demand is temperature-driven (Heating Degree Days, HDD). The seasonal index is modelled as:

$$\text{Seasonal}(m) = 1 + 0.35 \cos\left(\frac{(m-1) \cdot 2\pi}{12}\right)$$

with $m \in \{1, \ldots, 12\}$ indexing months. This generates the winter premium embedded in the forward curve.

#### 10.2.3. Fast Mean-Reversion

The high $\kappa = 2.10$ $\text{yr}^{-1}$ (half-life $\approx$ 4 months) reflects the fact that gas price spikes are rapidly corrected as LNG imports substitute for pipeline shortfalls. The term structure of gas futures is typically steep in backwardation during winter and flattens by summer, consistent with the model.

### 10.3. Uranio U3O8

#### 10.3.1. OTC Market

Uranium is not exchange-traded. Prices are set via bilateral long-term contracts (utilities and miners) and a thin spot market, leading to low liquidity and wide bid-ask spreads. The Sprott Physical Uranium Trust (URA ETF) is used as a market proxy.

#### 10.3.2. Slow Mean-Reversion

With $\kappa = 0.30$ $\text{yr}^{-1}$ (half-life $\approx$ 28 months), uranium prices respond slowly to supply/demand shocks. The 2021–24 price appreciation from $\approx$ \$30/lb to $>$ \$100/lb reflects a multi-year structural shift driven by: (a) renewed interest in nuclear power post-2022 energy crisis, (b) Kazatomprom production cuts (Kazakhstan supplies 43% of global primary production), and (c) financial demand via physical uranium trusts.

## 11. Weather Indicators and Integration into the Pricing Model

### 11.1. Motivation: Weather as a Supply and Demand Driver

Physical commodity prices are directly conditioned by meteorological variables. Unlike financial assets, whose supply and demand are primarily determined by macroeconomic variables, physical commodities have production and consumption cycles with strong climatic dependence.

QCEX integrates real weather data from the **Open-Meteo** free API (`archive-api.open-meteo.com`), which provides daily historical temperature and precipitation data without requiring an API key. When the connection is unavailable, the system generates synthetic data using sinusoidal functions calibrated to the historical seasonal patterns of each geographic zone.

## 11.2. Cocoa: ENSO and West African Rainfall

### 11.2.1. Transmission Mechanism

World cocoa production is concentrated in a narrow tropical belt. Ivory Coast and Ghana together produce 65% of global supply, making the market structurally vulnerable to any meteorological disruption in that region.

The main climate driver is the **ENSO** (*El Niño-Southern Oscillation*) phenomenon. El Niño episodes displace the intertropical convergence zone, reducing precipitation in West Africa during the critical months of flowering and pod formation (February–April). The causal chain is:

*El Niño → Reduced rainfall in Ghana/Ivory Coast → Water stress in plantations → Lower yield per tree → Reduced supply → Price increase*

The 2023–24 episode perfectly illustrates this mechanism: the most severe El Niño since 1998 caused a 25% production decline and drove prices from \$3,000/MT to over \$12,000/MT in less than twelve months.

### 11.2.2. Indicators Implemented

The `data/weather.py` module computes the following indicators for the reference location (Abidjan, 5.35°N / 4.00°W):

| **Indicator** | **Calculation** | **Bullish signal for price** |
|---|---|---|
| Precipitation 30d | Mean daily rainfall (mm) | $<$ historical mean |
| Dry days (30d) | Days with rainfall $< 0.5$ mm | $> 15$ days |
| Temperature anomaly | $\bar{T}_{30d} - \bar{T}_{\text{hist}}$ | $> +1.5$°C |
| ENSO proxy | $(\bar{P}_{30d} - \bar{P}_{\text{hist}})/\bar{P}_{\text{hist}}$ | $< -0.20$ |

Table 9: Weather indicators for Cocoa

### 11.2.3. Integration into the Pricing Model

Meteorological indicators can be incorporated as regressors in the long-term factor $\xi_t$ of the Schwartz-Smith model. The proposed extension is:

$$d\xi_t = \left(\mu_\xi + \beta_{\text{ENSO}} \cdot \text{ENSO}_t\right) dt + \sigma_\xi \, dW_\xi \tag{52}$$

where $\text{ENSO}_t$ is the proxy index calculated from the precipitation anomaly. A $\beta_{\text{ENSO}} < 0$ implies that dry periods (negative ENSO) generate upward pressure on the equilibrium price level.

## 11.3. Natural Gas TTF: Heating Degree Days (HDD)

### 11.3.1. Transmission Mechanism

Natural gas is the only commodity in the system with a dominant meteorological **demand** driver (unlike cocoa, whose driver is supply-side). Heating demand in Europe is directly correlated with outdoor temperature. The reference location is Berlin (52.52°N / 13.40°E), representative of the Central European market that determines TTF prices.

### 11.3.2. Heating Degree Days (HDD)

**Heating Degree Days** are the energy industry standard metric for quantifying thermal demand:

$$\mathrm{HDD}_d = \max\big(T_{\text{base}} - T_d,\ 0\big) \tag{53}$$

where $T_{\text{base}} = 15.5°\text{C}$ is the temperature below which heating activates, and $T_d$ is the daily mean temperature. The 30-day cumulative HDD:

$$\mathrm{HDD}_{30} = \sum_{d=t-29}^{t} \max(15.5 - T_d,\ 0) \tag{54}$$

The anomaly relative to the historical mean determines whether winter is colder or warmer than usual, generating respectively upward or downward pressure on TTF.

### 11.3.3. Sinusoidal Fallback

When real data is unavailable, the system approximates daily temperature in Berlin with:

$$T(m) = 12 + 10\cos\left(\frac{(m-1)\cdot 2\pi}{12}\right) + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0,4) \tag{55}$$

This sinusoidal function captures the annual cycle: minimum temperature in January ($T \approx 2°\text{C}$) and maximum in July ($T \approx 22°\text{C}$), with daily volatility of 2°C. **This is not a forecast** — it is a fallback to ensure the application functions without an internet connection.

## 11.4. Uranium U3O8: No Direct Weather Driver

Uranium is the only commodity in the system without direct weather dependence. Its prices are driven by:

1. **Nuclear policy**: number of reactors in operation and under construction
2. **Supply concentration**: Kazakhstan (Kazatomprom) controls 43% of global production
3. **Financial demand**: physical funds such as the Sprott Uranium Trust
4. **Long-term contract cycles**: utilities sign 5–10 year supply contracts, decoupling spot prices from short-term equilibrium

The uranium indicators panel in QCEX replaces meteorological variables with energy policy and market structure indicators, which are the real price drivers.

### 11.5. Module Architecture: `data/weather.py`

The module follows a two-tier architecture:

**T1. Real data**: HTTP request to the Open-Meteo API with a 5-second timeout. If the response is valid, indicators are computed on real historical data. The panel displays the `OPEN-METEO` label.

**T2. Synthetic fallback**: if the request fails (no internet, timeout, etc.), data is generated via the seasonal stochastic process described above. The panel displays the `SYNTHETIC` label.

This design ensures the application **always runs**, both in connected environments (real data) and in offline demonstrations (synthetic data).

## 12. Backtesting Engine

### 12.1. Motivation and Design Philosophy

A pricing model without empirical validation is incomplete. The QCEX backtesting engine tests whether systematic trading strategies based on the models implemented in the preceding sections generate positive risk-adjusted returns on historical data.

The engine is implemented in `analytics/backtest.py` and follows three principles:

1. **No look-ahead bias**: all signals are shifted by one day before computing returns, ensuring the strategy cannot use information not available at decision time.
2. **Transaction costs**: a configurable one-way cost (default 10 bps) is deducted on every position change.
3. **Benchmark comparison**: all strategies are compared against a buy-and-hold benchmark to measure genuine alpha generation.

### 12.2. Strategies Implemented

#### 12.2.1. Trend Following (EMA Crossover)

$$\text{Signal}_t = \begin{cases} +1 & \text{if } \text{EMA}_{20}(P_t) > \text{EMA}_{60}(P_t) \\ -1 & \text{if } \text{EMA}_{20}(P_t) < \text{EMA}_{60}(P_t) \end{cases} \tag{56}$$

The fast (20-day) EMA crossing above the slow (60-day) EMA generates a long signal. This strategy works well in commodities with persistent trends driven by structural supply or demand shifts.

#### 12.2.2. Mean Reversion (Bollinger Bands)

$$\text{Signal}_t = \begin{cases} +1 & \text{if } P_t < \mu_t - 1.5\,\sigma_t \quad \text{(oversold)} \\ -1 & \text{if } P_t > \mu_t + 1.5\,\sigma_t \quad \text{(overbought)} \\ 0 & \text{otherwise} \end{cases} \tag{57}$$

where $\mu_t$ and $\sigma_t$ are the 20-day rolling mean and standard deviation. This strategy is effective for commodities that exhibit strong mean-reversion behaviour, such as natural gas, where price spikes are rapidly corrected by supply responses.

#### 12.2.3. Momentum

$$\text{Signal}_t = \text{sign}\left(\frac{P_t - P_{t-21}}{P_{t-21}}\right) \tag{58}$$

One-month price momentum. Goes long on assets with positive recent returns, short on negative. Works well when price trends persist beyond the measurement window.

#### 12.2.4. Weather-Driven (Seasonal)

A rules-based strategy using the seasonal patterns derived in Section 11:

- **Cocoa**: long in April–June (post-dry season, supply uncertainty); short in October–December (main harvest peak)
- **Gas TTF**: long in October–February (winter heating season); short in April–August (injection season, low demand)
- **Uranium**: flat (no seasonal driver)

#### 12.2.5. Volatility Breakout

$$\text{Signal}_t = \begin{cases} +1 & \text{if } P_t > \max_{20}(P) + 1.5 \cdot \text{ATR}_{20} \\ -1 & \text{if } P_t < \min_{20}(P) \\ 0 & \text{otherwise} \end{cases} \tag{59}$$

where $\text{ATR}_{20}$ is the 20-day average true range. Captures breakouts from consolidation phases.

#### 12.2.6. Combined (Equal-Weight Ensemble)

$$\text{Signal}_t^{\text{combined}} = \text{clip}\left(\frac{1}{5}\sum_{i=1}^{5} \text{Signal}_t^{(i)},\ -1,\ +1\right) \tag{60}$$

An equal-weight combination of all five strategies. By diversifying across uncorrelated signal types, the ensemble typically reduces drawdown while maintaining positive returns.

### 12.3. Performance Metrics

| Metric | Formula | Interpretation |
|---|---|---|
| Annualised Return | $\bar{r} \times 252$ | Average yearly return |
| Sharpe Ratio | $(\bar{r} - r_f)/\sigma \times \sqrt{252}$ | Return per unit of total risk |
| Sortino Ratio | $(\bar{r} - r_f)/\sigma_{\text{down}} \times \sqrt{252}$ | Return per unit of downside risk |
| Max Drawdown | $\min_t \frac{\text{NAV}_t - \max_{s<t} \text{NAV}_s}{\text{NAV}_0}$ | Worst peak-to-trough loss |
| Calmar Ratio | Ann. Return / \|Max Drawdown\| | Return relative to worst loss |
| Win Rate | $\frac{\#\{r_t > 0\}}{T} \times 100$ | Fraction of profitable days |

Table 10: Performance metrics computed by the backtesting engine

### 12.4. Empirical Results

#### 12.4.1. Gas Natural TTF (2 years)

The most striking result is that **Mean Reversion is the best-performing strategy** for Gas TTF, achieving a Sharpe ratio of 0.56 and a total return of +47% over two years, while the buy-and-hold benchmark is in negative territory (gas prices declined from their 2022–23 crisis highs).

This is economically intuitive: natural gas prices exhibit strong mean-reversion because supply (LNG imports, storage injections) responds relatively quickly to price spikes. Trend-following strategies lose money because they chase rallies that subsequently reverse.

**Key insight**: the optimal strategy depends on the commodity's microstructure. Mean-reverting commodities (gas, power) favour contrarian strategies. Structural supply-shock commodities (cocoa 2023–24) may favour trend-following.

#### 12.4.2. Important Caveats

**C1. Data quality**: price data comes from ETF proxies (NG=F, CC=F, URA) rather than exchange-traded futures chains. Roll costs and basis risk are not captured.

**C2. Overfitting risk**: with only 5 strategies and 2 years of data, results may not be statistically significant. A minimum of 10 years and 200+ independent trades is recommended for robust inference.

**C3. Market impact**: the backtest assumes trades execute at close prices with fixed 10 bps cost. Real execution on illiquid commodities (uranium) would face significantly higher costs and slippage.

**C4. Regime changes**: the 2022–23 energy crisis represents a structural break. Strategies calibrated on post-crisis data may not generalise to other regimes.

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**Pablo M. Paniagua** · QCEX — Quantum Commodity Exchange

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